

From Risk Models to AI Models

Governing the Next Horizon in Finance

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I recently spoke at the 2025 Philippine Investment Conference organized by the CFA Society Philippines. My talk was on risk and governance—moving from the financial models we know, like Expected Credit Loss and Value-at-Risk, into the world of AI models. The audience was mostly investment professionals, but the lessons aren't limited to finance. They're about how we think about innovation, risk, and the governance structures that keep tools from becoming weapons. Here's an extended version of that talk, reworked for Signal & Noise.

Finance has always been a proving ground for innovation.¹ Derivatives allowed investors to hedge and speculate. They unlocked precision but also magnified exposures when poorly monitored.² Value-at-Risk (*VaR*) promised a clean measure of exposure. History shows, though, that breakthroughs become fault lines when oversight fails. [1-2] Long-Term Capital Management, run by Nobel laureates and top financiers, collapsed when leverage met extreme market shifts.³ [3-4] The math worked in ordinary times, but the firm had little margin for edge cases.⁴ American International Group (AIG) nearly brought down the system when collateral calls overwhelmed its credit default swaps. [5-6] *VaR*,⁵ treated as a precise forecast rather than a probabilistic estimate, lulled boards and regulators into complacency. [7-8]

In each case, the models did what they were built to do⁶; i.e., they summarized risk and quantified exposures. The failure was governance. Leaders mistook model results for facts, when they were meant to be guides. Regulators also moved too slowly. Model innovation did not destabilize finance; weak oversight did.

Now comes a new wave of innovation: artificial intelligence (AI). AI models bring clear opportunities: better decisions, sharper insights, new ways to compete. But it also carries risks that are only beginning to be understood: bias, opacity, and systemic effects that ripple across institutions. As with past innovations, the danger is less the technology or model itself but the delay in governance. The real question is whether oversight can catch up before overconfidence sets the stage for the next crisis.

Familiar ground, new terrain

To understand what's new about AI models, it helps to start with what is familiar. Consider the Expected Credit Loss (*ECL*) framework. These models forecast loan impairments using historical data and macroeconomic assumptions. Because they directly shape balance sheets and capital provisions, they demand rigorous validation, backtesting, and documentation.⁷ Banks maintain entire teams just to ensure model assumptions are sound, parameters are calibrated, and results can withstand external audit.⁸

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1. Financial markets often absorb new tools earlier than other industries because the incentives are immediate and measurable.
2. Derivatives themselves aren't villains. They are just tools. The real issue is that layers of contracts made the system opaque, so nobody could see the whole web until it was already tangled.
3. LTCM's collapse is still studied in risk management programs as a cautionary tale of overconfidence in quantitative models.
4. From a complexity view, LTCM tried to tame a nonlinear system with linear tools. Once shocks cascaded, their models had no way to adapt.
5. That number was easy to communicate to boards and regulators. Because it appeared precise and standardized, many institutions began relying on it as the main measure of risk, often without supplementing it with stress tests or scenario analysis.
6. One could also argue these models were oversimplified. By compressing a complex, adaptive financial system into a few metrics, they created a false sense of control. Complexity science tells us that systems with feedback loops and emergent behavior resist neat modeling; reducing them to single numbers invites fragility rather than resilience.

7. Regulatory bodies, such as central banks, often scrutinize *ECL* models closely, challenging banks on their robustness, data inputs, and economic assumptions.
8. This illustrates how much institutional effort already goes into governing even transparent, traditional models.

VaR, meanwhile, gave finance a standardized way to measure and communicate risk. But experience showed its limitations. Before 2008, many firms leaned too heavily on VaR, underestimating tail risks. When markets moved outside those assumptions,⁹ losses were severe. The lesson was that *VaR* needed stress testing, challenge, and supplementation with judgment.

These cases show that even transparent, interpretable models require extensive oversight. Independent validation, continuous monitoring, stress scenarios, and careful documentation are essential. Neither operates as a black box¹⁰; their assumptions and mechanics can be explained. Even then, the industry still demands layers of oversight and control, because without guardrails these models can mislead decision-makers with a false sense of certainty.

Now compare that to AI models. They sit on this same continuum but raise the stakes. Credit scoring engines, fraud detection algorithms, and portfolio optimization systems also forecast, classify, and influence provisioning. While some AI methods, like linear models and decision trees, remain interpretable, the “black box” concern applies mostly to complex architectures like deep neural networks.¹¹ These systems introduce a far broader risk surface. Their assumptions are often implicit, buried in training data; parameters are counted in the millions, and more.¹²

Algorithmic innovations make AI systems harder to interpret. Errors can’t always be traced to a single assumption or parameter. Governance, in turn, has to go beyond backtesting or validation; it has to also look into bias detection, lifecycle monitoring, explainability¹³, adversarial resilience, and ethical accountability. If it takes whole governance infrastructures to keep interpretable models like *ECL* and *VaR* under control, then AI demands at least as much rigor, if not more.¹⁴

Mapping AI harms in risk categories

One way to bring AI into the language of finance is to map its harms into familiar risk categories. In the conference slides (*slide 9*), we listed common AI harms and showed how each could create **financial risk**, **operational risk**, **ethical/reputational risk**, and **systemic risk**.¹⁵ Looking at harms through these categories makes clear why governance has to be multi-dimensional.

AI Harm	Financial Risk	Operational Risk	Ethical / Reputational Risk	Systemic Risk
Cost	High cost of failure → financial losses	Inefficient resource use	Perception of waste, mismanagement	If industry-wide, undermines efficiency of adoption
Bias	Mispriced credit, unfair asset allocation	Weak data/process controls	Discrimination, reputational damage	Widespread bias → loss of trust in financial markets
False Negatives	Missed fraud → direct losses	Detection/control failures	Perceived negligence	If sector-wide, systemic fraud risk
Black Box / Transparency	Poor risk pricing, regulatory fines	Inability to audit models	Erodes investor / regulator trust	Market-wide opacity → contagion
Adversarial Attacks	Direct trading / manipulation losses	Cybersecurity gaps	Trust breached (investors, clients)	Potential for coordinated systemic attacks
Model Drift	Under/over-provisioning, mispriced risk	Failure to recalibrate models	Perception of negligence	If common across firms, distorted credit or market signals
Accuracy	Mispricing assets, wrong forecasts	Poor model performance	Investor mistrust	If replicated widely, systemic misallocation of capital
Hallucination	False disclosures → fines, mispriced assets	Weak validation and review controls	Loss of credibility with regulators/clients	If widely trusted, systemic misallocation of capital

The table makes another point clear: these harms rarely stay confined to one category. They spill over, with a single weakness showing up as multiple risks.

9. *VaR* is like summarizing the weather with “20% chance of rain” and forgetting storms exist. Useful, but dangerously incomplete when treated as the whole story.

10. Black box refers to a system whose internal workings are not easily interpretable. It’s when a model does not or cannot tell you how it reached its decisions.

11. Why use deep neural networks then? Because they can pick up patterns in truly complex datasets that simpler models would miss. Think image recognition, language translation, or spotting subtle fraud signals in millions of transactions. The trade-off is power for interpretability; you get more accuracy on hard problems, but at the cost of transparency... for now.

12. Artificial neural networks, with millions to billions of parameters, are especially sensitive to these design choices; small changes can shift outcomes significantly.

13. Research in explainable AI (XAI) is advancing, aiming to make neural networks more interpretable through methods like feature attribution and surrogate models.

14. Until explainability becomes standard, governance must lean heavily on testing, monitoring, and escalation protocols.

15. These categories mirror the way financial institutions already structure risk management. Financial risks affect balance sheets directly, operational risks point to process failures, ethical and reputational risks erode trust, and systemic risks spread across markets. Framing AI harms this way makes them easier to integrate into existing governance and oversight frameworks.

Take *model drift*¹⁶. In credit scoring, when it's wrong, errors are more than just the actual technical failure; it could misprice loans (financial), reveal weak monitoring (operational), look like negligence to regulators (reputational), and if replicated across firms, could distort entire credit markets (systemic).

A more recent risk, *hallucinations*¹⁷, is a particular concern tied with generative AI. A hallucinated due diligence report or legal memo, for instance, could lead to false disclosures and regulatory penalties. That creates financial risk through fines, operational risk through failed validation, and reputational risk through lost credibility. If many institutions rely on similar outputs, hallucinations could drive systemic misallocation of capital.

A hypothetical credit scoring scenario

The harms discussed in the previous section become sharper when viewed through a specific use case. Credit scoring is a sharp lens for both the promise and risk of AI. [9] [10] AI-driven credit scoring systems can process vast datasets beyond traditional credit scores, such as transaction histories, digital footprints, even behavioral patterns. The results are compelling with faster loan approvals, lower default rates, and improved predictive accuracy. For banks, profitability rises, non-performing loans fall, and customer service speeds up.

But problems can emerge. Imagine a model that generalizes risk across places with high exposure to natural hazards. (*slides 13-14*). Loan applicants from Samar and Leyte,¹⁸ for instance, consistently receive lower scores than traditional models would suggest. Even individuals with strong repayment histories are penalized.¹⁹ The system, while profitable overall, embeds geographic and demographic bias.²⁰

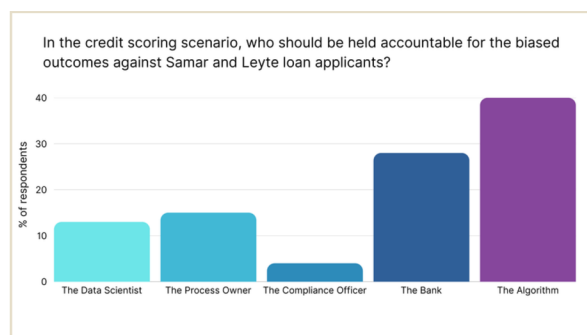
The hidden costs²¹ are significant:

- **Financial:** potential ₱150M in lost lending opportunities annually.
- **Regulatory:** up to ₱50M in fines for discrimination and financial exclusion.²²
- **Reputational:** a 3–5% decline in market share in the affected regions

This is the paradox of AI in finance: a model can “work” in the aggregate while quietly undermining entire communities. And in one sense, the model may not even be entirely wrong. Models generalize from the data they're given. The real question is whether risk officers and accountable leaders actually understand how the model works, what it is optimizing for, and what it may be sacrificing. Profitability does not have to come at the cost of exclusion, but preventing that outcome requires awareness and active oversight.

Who Is Accountable?

The credit scoring scenario raises a deeper question: “*when harms emerge, who is accountable?*” In a live session with more than 100 compliance officers, the Samar and Leyte case was presented, and participants were asked directly who should take responsibility for the biased outcomes.



16. Model drift refers to the degradation of a model's accuracy when the data it was trained on no longer reflects current realities.

17. In their current state, large language models (LLMs) do not “understand” content in a human sense. They generate text by predicting the most likely sequence of words based on statistical patterns in training data. This probabilistic process explains why they can produce fluent but made-up outputs that are commonly called *hallucinations*.

18. Samar and Leyte are provinces in the Philippines that sit along the country's typhoon belt and are regularly exposed to strong storms, flooding, and other natural hazards. This frequent exposure makes them a common reference point in disaster risk and climate-related studies.

19. Algorithmic bias arises when group-level generalizations override individual-level evidence.

20. On the other hand, research also points out that AI applications in finance could also expand inclusion by addressing challenges faced by unbanked and underbanked populations. [11]

21. These numbers are hypothetical, meant only to illustrate scale.

22. Under the Financial Products and Services Consumer Protection Act (Republic Act 11765), implemented through BSP Circular No. 1048 (2019), banks must treat clients fairly and avoid discrimination. Breaches may lead to sanctions and fines. As noted in sidenote 18, the figure cited here is hypothetical and for illustration only.

Survey²³ results showed a telling split in the room. Some pointed to the data scientist, others the compliance officer, some the board or the vendor—and a large share pointed to the algorithm itself (*slide 16*).²⁴

That distribution highlights the governance challenge. In practice, many still instinctively look to “the algorithm” as if it could bear responsibility. But governance cannot allow that escape hatch. Accountability must remain with people and institutions, because only they can enforce standards, run bias audits, and recalibrate when things drift.

From principles to practice

So what now? Yes, accountability is essential, but it requires infrastructure to support it. Many institutions struggle here. One common problem is treating AI principles, which are broad statements of intent, as if they were governance. Principles set direction; governance provides the processes, controls, and accountability to make them real.

Principles are values: fairness, transparency, accountability.²⁵ They set the ethical compass, but governance is what translates them into practice. Governance translates values into enforceable policies, builds processes that guide AI teams, and sets controls that detect and correct errors. It establishes documentation, monitoring, and clear lines of responsibility.

Many statements of principle on AI already exist, but three are worth highlighting for this discussion²⁶:

- **Pro-innovation mindset.** Governance should not stifle experimentation.²⁷ Done well, it builds clarity and trust, allowing institutions to deploy AI faster, not slower.
- **Risk-centric.** Scrutiny should scale with the level of risk.²⁸ A chatbot for FAQs should not be governed like a credit scoring engine.
- **Principle of proportionality.** Oversight should be proportionate to both complexity and sensitivity.²⁹ Low-risk projects avoid unnecessary bureaucracy, while high-risk systems face rigorous checks.

These principles sound right in theory, but they immediately raise the harder question: how are they operationalized? It is one thing to declare a pro-innovation mindset, risk-centricity, and proportionality; it is another to build the structures, processes, and roles that make them work in practice. That takes us to the next step: operationalizing AI governance.

Operationalizing AI governance

The ECAIR AI Governance Framework³⁰ provides a structured way to move from principle to practice. It is designed for institutions that are already building and deploying high-risk AI systems, where governance cannot remain abstract. It has to be implemented in daily operations.

Its central idea is that **governance must produce artifacts at every stage of the AI development lifecycle**. These are the documents, records, and decisions that show what was done, why it was done, and who was accountable. Without these, governance remains aspirational. With them, it becomes auditable and enforceable.

23. The survey was done live using Menti, which made the division in the room immediately visible.
24. To be fair, this is not a straightforward question since responsibility in AI projects is distributed across technical and business teams. Still, the responses were surprising. If the same question had been asked about an *ECL* model failing, few would have blamed the model itself.
25. International bodies have published widely recognized AI principles. The OECD AI Principles (2019) highlight inclusive growth, human-centered values, transparency, robustness, and accountability. The UN has echoed similar themes, emphasizing fairness, privacy, safety, and sustainable development.
26. These are the three I emphasized during my talk at the Philippine Investment Conference with CFA and other finance professionals. There are many other principles out there, but I focused on these because they are the most actionable in a finance setting and allow for deeper discussion, rather than a long list that stays abstract.
27. Modern AI techniques like transformers advance quickly; governance should enable safe piloting rather than block it.
28. Deep learning models used in credit scoring or medical triage carry far higher stakes than those used in recommending the next movie on Netflix.
29. This mirrors financial regulation, where the complexity of an instrument drives supervisory intensity.
30. This is the framework I emphasized during the Philippine Investment Conference. As Managing Director and Chief AI and Data Officer at ECAIR, I live by this framework in practice; we implement it day to day, especially in building high-risk AI systems.

The framework is implemented in three main ways:

- **Three lines of defense.** Data scientists and project officers manage risks and document lifecycle artifacts. A governance council reviews, escalates, and makes go/no-go decisions (*slides 26–29*). Responsible AI is not the work of one role; it is intentionally distributed across functions.
- **Lifecycle controls.** Governance is embedded at every stage: plan, design, develop, deploy, monitor.³¹ Each stage requires outputs: impact assessments, validation reports, monitoring logs, and escalation records.³² This makes governance traceable, not just declared (*slides 30–31*).
- **Alignment with global standards.** The framework aligns with ISO/IEC 42001, 42005, 23894, IEEE 7010, and related standards (*slides 24–25*). This ensures that governance practices are not only internal but benchmarked against global best practice.

31. Lifecycle governance mirrors software development practices but adds compliance artifacts at each phase.
32. These artifacts serve as evidence for regulators and boards that AI risks are actively managed.



The ECAIR model emphasizes that governance is built in, not bolted on. It is not an extra layer of paperwork after a model goes live, but rather a process that produces evidence of responsible design, validation, and monitoring at each step.

This mindset should feel familiar to finance. Risk officers already know what it takes to govern models like *ECL* and *VaR*: documentation, validation, independent review, and regulatory oversight. The difference with AI is scale, opacity, and a broader surface for harm. Yet the principle is the same. Just as *VaR* and *ECL* required governance infrastructures to protect markets, AI systems require even stronger governance to safeguard trust and stability.

For risk teams, this means evolving their capabilities. As institutions mature, reliance on external vendors or surface-level validations won't suffice. Risk functions will need data scientists and AI engineers within their ranks, able to audit algorithms directly. The ability to probe models, test assumptions, and challenge outputs should become a core part of risk governance, much like credit analysts and quants became integral in earlier eras.³³

The Future Still Hinges on Governance

Finance has long been a proving ground for innovation. The lesson across history is that governance determines whether new tools stabilize markets or destabilize them.

AI governance is often seen as bureaucracy, slowing innovation. In practice, it is what keeps innovation aligned with responsibility and market stability. Without it, AI can push institutions off course. With it, organizations can innovate faster and more safely.

ECL and *VaR* already showed the role of governance in shaping outcomes. AI brings the same challenge—only at greater scale. The future of finance will hinge less on who adopts AI first, and more on who governs it well.

33. I expect that in a few years, having AI scientists inside risk teams will be as normal as having quants is today. Without that in-house expertise, boards will always be one step behind the algorithms they're supposed to oversee.

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